

Module 7. Neural Networks

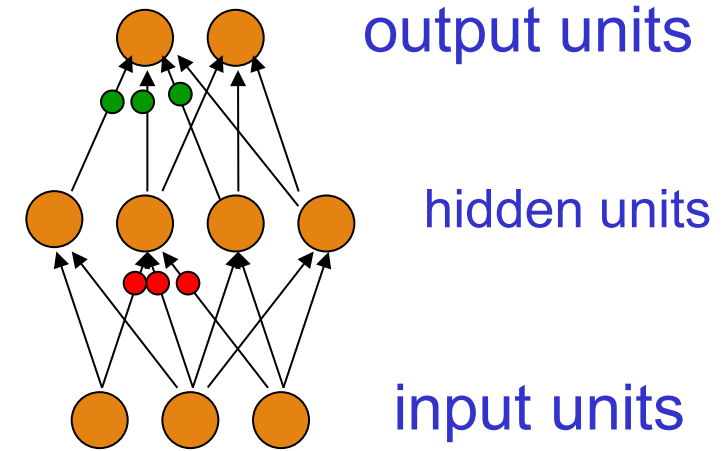
- ❑ Basic Concepts
- ❑ Perceptron Learning
- ❑ Artificial Neural Networks
- ❑ Backpropagation 
- ❑ Summary

Efficient way to learn the weights

- ❑ Networks without hidden units are very limited in the input-output mappings they can model.
 - ❑ More layers of linear units do not help. Its still linear.
 - ❑ Fixed output non-linearities are not enough
- ❑ We need multiple layers of adaptive non-linear hidden units. This gives us a universal approximator. But how can we train such nets?
 - ❑ We need an efficient way of adapting **all** the weights, not just the last layer. This is hard. Learning the weights going into hidden units is equivalent to learning features.
 - ❑ Nobody is telling us directly what hidden units should do.

Naïve Approach: Learning by perturbing weights

- ❑ Randomly perturb one weight and see if it improves performance. If so, save the change.
- ❑ **Very inefficient.** We need to do multiple forward passes on a representative set of training data just to change one weight.
- ❑ Towards the end of learning, large weight perturbations will nearly always make things **worse**.
- ❑ We could randomly perturb all the weights in parallel and correlate the performance gain with the weight changes.
- ❑ Not any better because we need lots of trials to “see” the effect of changing one weight through the noise created by all the others.



- ❑ Learning the hidden to output weights is **easy**.
- ❑ Learning the input to hidden weights is **hard**.

The idea behind back propagation

- ❑ We don't know what the hidden units ought to do, but we can compute how fast the error changes as we change a hidden activity.
- ❑ Instead of using desired activities to train the hidden units, use **error derivatives w.r.t. hidden activities**.
- ❑ Each hidden activity can affect many output units and can therefore have many separate effects on the error. These effects must be combined.
- ❑ We can compute error derivatives for **all** the hidden units efficiently.
- ❑ Once we have the error derivatives for the hidden activities, its easy to get the error derivatives for the weights going into a hidden unit.